浙江大学2019-20秋冬《数据结构基础》期中模拟练习（45分钟）

开始时间 01/01/2016 8:00:00 AM **结束时间** 01/18/2038 8:00:00 AM **答题时长** 45分钟
考生 LiXiao **得分** 78 **总分** 100

判断题

得分：12 总分：15

1-1 In a directed graph, the sum of the in-degrees must be equal to the sum of the out-degrees of all the vertices. (3分)

(•) T ( ) F

评测结果 答案正确（3 分）

1-2 If a linear list is represented by a linked list, the addresses of the elements in the memory must be consecutive. (3分)

( ) T (•) F

评测结果 答案正确（3 分）

1-3 ADT is the abbreviation for Abstract Data Type in the textbook of data structures. (3分)

(•) T ( ) F

评测结果 答案正确（3 分）

1-4 In *Union/Find* algorithm, if *Unions* are done by size, the depth of any node must be no more than $N/2$, but not $O(logN)$. (3分)

( ) T (•) F

评测结果 答案正确（3 分）

1-5 N is $O(\sqrt{N}logN)$. (3分)

(•) T ( ) F

评测结果 答案错误（0 分）

单选题

得分：60 总分：60

2-1 For a sequentially stored linear list of length N, the time complexities for query and insertion are: (5分)

A. $O(1), O(1)$
B. $O(1), O(N)$
C. $O(N), O(1)$
D. $O(N), O(N)$

评测结果 答案正确（5 分）

2-2 If a binary search tree of N nodes is complete, which one of the following statements is FALSE? (5分)

A. the average search time for all nodes is $O(logN)$
B. the minimum key must be at a leaf node
C. the maximum key must be at a leaf node
D. the median node must either be the root or in the left subtree

评测结果 答案正确（5 分）

2-3 Let n be a non-negative integer representing the size of input. The time complexity of the following piece of code is: (5分)

```
x = 0;
while ( n >= (x+1)*(x+1) )
    x = x+1;
```

A. $O(logn)$
B. $O(n^{1/2})$
C. $O(n)$
D. $O(n^2)$

评测结果 答案正确（5 分）

2-4 The array representation of the disjoint sets is given by { 2, −4, 2, 2, −3, 5, 6, 9, −2 }. Keep in mind that the elements are numbered from 1 to 9. After invoking Union(Find(7), Find(9)) with union-by-size, which elements will be changed in the resulting array? (4分)

A. elements of index 5 and 7
B. elements of index 7 and 9
C. elements of index 5 and 9
D. elements of index 5, 7 and 9

评测结果 答案正确（4 分）

2-5 Since the speed of a printer cannot match the speed of a computer, a buffer is designed to temperarily store the data from a computer so that later the printer can retrieve data in order. Then the proper structure of the buffer shall be a: (5分)

A. stack

B. queue
C. tree
D. graph

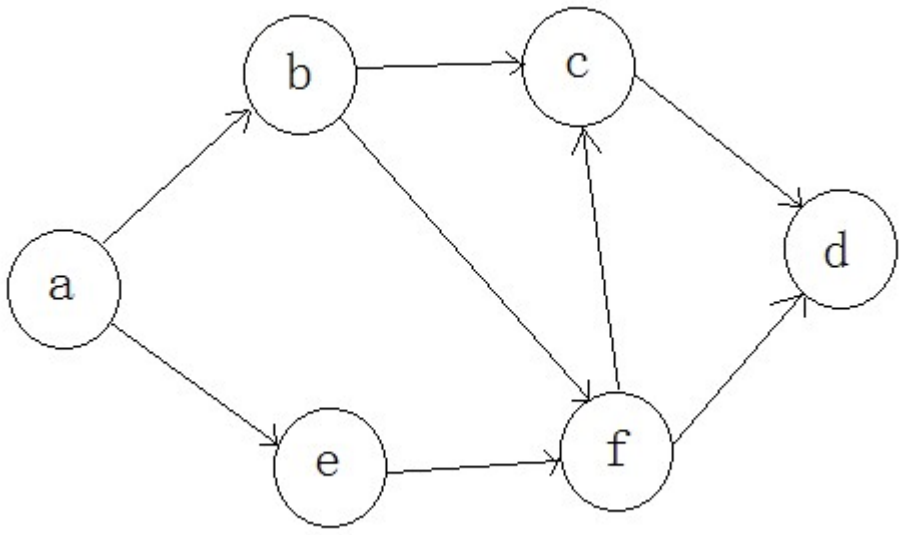
评测结果 答案正确（5 分）

2-6 Given the popping sequence of a stack as {a, b, c, d, e}. Among the following, the impossible pushing sequence is: (5分)

A. c b a e d
B. d e a c b
C. e a b c d
D. e d a b c

评测结果 答案正确（5 分）

2-7 How many distinct topological sequences are there in the following DAG? (5分)



A. 2
B. 3
C. 4
D. 5

评测结果 答案正确（5 分）

2-8 If graph G is NOT connected and has 20 edges, then it must have at least __ vertices. (5分)

A. 7
B. 8
C. 9
D. 10

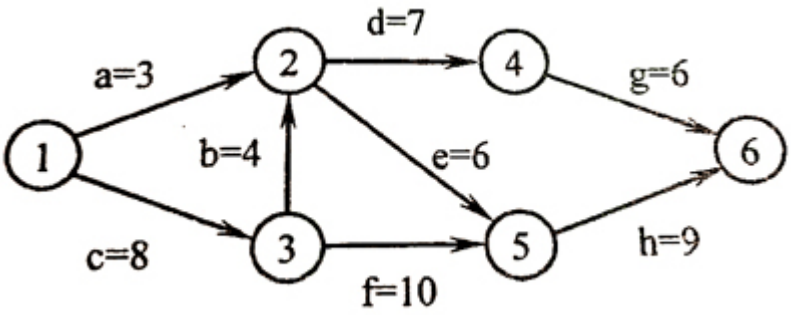
评测结果 答案正确（5 分）

2-9 A full tree of degree 3 is a tree in which every node other than the leaves has 3 children. How many leaves does a full tree of degree 3 have if it has 217 nodes? (5分)

- A. 71
- B. 72
- C. 145
- D. 146

评测结果 答案正确（5 分）

2-10 The following figure shows the AOE network of a project with 8 activities. The earliest and the latest completion times of the activity `d` are __, respectively. (5分)



- A. 3 and 7
- B. 12 and 12
- C. 12 and 14
- D. 15 and 15

评测结果 答案正确（5 分）

2-11 If besides finding the shortest path from `S` to every other vertices, we also need to count the number of different shortest paths, we can modify the Dijkstra algorithm in the following way: add an array `count[]` so that `count[V]` records the number of different shortest paths from `S` to `V`. Then `count[V]` shall be initialized as: (5分)

- A. `count[S]=1;` and `count[V]=0` for other `V`
- B. `count[S]=0;` and `count[V]=1` for other `V`
- C. `count[V]=1` for all vertices
- D. `count[V]=0` for all vertices

评测结果 答案正确（5 分）

2-12

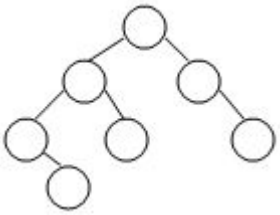


Which one of the following is the data structure that is best represented by the above picture? (1分)

- A. Circularly linked list
- B. Directed graph
- C. Two-dimensional array
- D. Complete binary tree

评测结果 答案正确（1 分）

2-13 Given the shape of a binary tree shown by the figure below. If its inorder traversal sequence is { `e`, `a`, `c`, `b`, `d`, `g`, `f` }, then the node on the same level of `c` must be: (5分)



- A. `a`
- B. `g`
- C. `d`
- D. None

评测结果 答案正确（5 分）

程序填空题

得分：6 总分：25

5-1 Please fill in the blanks in the program which deletes a given element at position `p` from a min-heap `H`.

```
Deletion ( PriorityQueue H,  int p )  /* delete the element H->Elements[p] */
{
   ElementType temp;
   int child;
```

```
    temp = H-> Elements[ H->Size-- ];
    if ( temp < H->Elements[p] ) {
        while ( (p != 1) && (temp < H->Elements[p/2]) ) {
            H[i/2]=H[i]                                    (3分);
            p /= 2;
        }
    }
    else {
        while( (child = 2*p) <= H->Size) {
            if ( child != H->Size &&                                    (3分) )
                child ++;
            if (  H->Elements[p] < H->Elements[child];                  (3分) ) {
                H->Elements[p] = H->Elements[child];
                p = child;
            }
            else
                break;
        }
    }
    H->Elements[p] = temp;
}
```

评测结果 答案错误（0 分）

测试点得分

序号	结果	得分
0	编译错误	0
1	未作答	0
2	编译错误	0

5-2 The function `Dijkstra` is to find the shortest path from `Vertex S` to every other vertices in a given `Graph`. The distances are stored in `dist[]`, and `path[]` records the paths. The `MGraph` is defined as the following:

```
typedef struct GNode *PtrToGNode;
struct GNode{
    int Nv;  /* Number of vertices */
    int Ne;  /* Number of edges    */
    WeightType G[MaxVertexNum][MaxVertexNum]; /* adjacency matrix */
};
typedef PtrToGNode MGraph;
```

```
void Dijkstra( MGraph Graph, int dist[], int path[], Vertex S )
{
    int collected[MaxVertexNum];
    Vertex V, W;

    for ( V=0; V<Graph->Nv; V++ ) {
        dist[V] = Graph->G[S][V];
```

```
        path[V] = -1;
        collected[V] = false;
    }
    dist[S] = 0;
    collected[S] = true;

    while (1) {
        V = FindMinDist( Graph, dist, collected );
        if ( V==ERROR ) break;
        collected[V] = true;
        for( W=0; W<Graph->Nv; W++ )
            if ( collected[W]==false && Graph->G[V][W]<INFINITY ) {
                if ( dist[W]>dist[v]+Graph->G[V][W]  (4分) ) {
                    dist[W] = dist[v]+Graph->G[V][W]  (3分);
                    path[W] = W  (3分);
                }
            }
    } /* end while */
}
```

评测结果 答案错误（0 分）

测试点得分

序号	结果	得分
0	编译错误	0
1	编译错误	0
2	答案错误	0

5-3 The function is to return the reverse linked list of `L`, with a dummy header.

```
List Reverse( List L )
{
    Position Old_head, New_head, Temp;
    New_head = NULL;
    Old_head = L->Next;

    while ( Old_head )  {
        Temp = Old_head->Next;
        Old_head->Next=New_head  (3分);
        New_head = Old_head;
        Old_head = Temp;
    }
    L->Next=New_head  (3分);
    return L;
}
```

评测结果 答案正确（6 分）

测试点得分

序号	结果	得分
0	答案正确	3

序号	结果	得分
1	答案正确	3